50 Linux Commands

File & Directory Commands :

1. pwd – print working directory

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ pwd
/home/ImmunIty
```

2. ls – list files

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
Task
```

3. ls -l – detailed listing

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls -l
total 4
drwxr-xr-x 2 ImmunIty ImmunIty 4096 Mar 23 00:33 Task
```

4. ls -a – show hidden files

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls -a
.   .bash_history   .bashrc           .config  .local    Task       .zshrc
..  .bash_logout    .bashrc.original  .java    .profile  .zprofile
```

5. cd – change directory

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ cd Task/

┌──(ImmunIty㉿1mmun1ty)-[~/Task]
└─$ |
```

6. cd .. – go back

```
┌──(ImmunIty㉿1mmun1ty)-[~/Task]
└─$ cd ..

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ |
```

7. cd /path – go to path

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ cd /usr/share/

┌──(ImmunIty㉿1mmun1ty)-[/usr/share]
└─$ |
```

8. mkdir newfolder – create folder

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ mkdir newfolder

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
newfolder  Task
```

9. rmdir folder – remove folder

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ rmdir newfolder/

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
Task
```

10. touch file.txt – create a file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ touch file.txt

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
file.txt  Task
```

11. cp file1 file2 – copy file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ cp file.txt file1.txt

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
file1.txt  file.txt  Task
```

12. mv old new – rename or move file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ mv file1.txt file2.txt

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
file2.txt  file.txt  Task
```

13. rm file – delete file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ rm file2.txt

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
file.txt  Task
```

14. rm -r folder – delete folder recursively

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ rm -r Task/

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
file.txt
```

15. cat file.txt – view file content

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ cat file.txt
I am ali
```

16. more file.txt – scroll file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ more scroll.txt
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
```

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ tail scroll.txt
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
```

20. tail -f file.log – follow log updates

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ tail -f scroll.txt
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
^Z
[3]+  Stopped                 tail -f scroll.txt
```

System & Info Commands

21. whoami

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ whoami
ImmunIty
```

22. hostname

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ hostname
1mmun1ty
```

23. date

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ date
Mon Mar 23 03:03:54 AM PKT 2026
```

24. uptime

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ uptime
 03:04:54 up  2:39,  1 user,  load average: 0.00, 0.00, 0.00
```

25. uname -a

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ uname -a
Linux 1mmun1ty 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 GNU/Linux
```

26. free -h

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ free -h
               total        used        free      shared  buff/cache   available
Mem:            11Gi       490Mi        11Gi       3.5Mi       160Mi        11Gi
Swap:          3.0Gi          0B       3.0Gi
```

27. df -h

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            5.8G     0  5.8G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            5.8G  4.0K  5.8G   1% /mnt/wsl
drivers         184G  160G   25G  87% /usr/lib/wsl/drivers
/dev/sdd       1007G  837M  955G   1% /
none            5.8G   92K  5.8G   1% /mnt/wslg
none            5.8G     0  5.8G   0% /usr/lib/wsl/lib
rootfs          5.8G  2.7M  5.8G   1% /init
none            5.8G     0  5.8G   0% /dev
none            5.8G  476K  5.8G   1% /run
none            5.8G     0  5.8G   0% /run/lock
none            5.8G     0  5.8G   0% /run/shm
none            5.8G   76K  5.8G   1% /mnt/wslg/versions.txt
none            5.8G   76K  5.8G   1% /mnt/wslg/doc
C:\             184G  160G   25G  87% /mnt/c
D:\             292G  200G   93G  69% /mnt/d
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/console-getty.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           1.2G  8.0K  1.2G   1% /run/user/1000
```

28. Top

```
top - 03:07:31 up  2:42,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  21 total,   1 running,  17 sleeping,   3 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  11850.2 total,  11358.9 free,    499.6 used,    160.5 buff/cache
MiB Swap:   3072.0 total,   3072.0 free,      0.0 used.  11350.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   23796  13880  10680 S   0.0   0.1   0:01.24 systemd
      2 root      20   0    3120   1920   1920 S   0.0   0.0   0:00.00 init-systemd(ka
      7 root      20   0    3120   1792   1792 S   0.0   0.0   0:00.00 init
     44 root      20   0   50072  14464  13312 S   0.0   0.1   0:00.25 systemd-journal
     72 root      20   0   33552  10880   8064 S   0.0   0.1   0:02.33 systemd-udevd
     80 root      20   0    4280   2432   2304 S   0.0   0.0   0:00.03 cron
     81 message+  20   0    7888   3840   3584 S   0.0   0.0   0:00.06 dbus-daemon
    114 root      20   0   17776   8064   7040 S   0.0   0.1   0:00.19 systemd-logind
    249 root      20   0    5168   2560   2432 S   0.0   0.0   0:00.01 agetty
    250 root      20   0    5124   2304   2176 S   0.0   0.0   0:00.01 agetty
    255 root      20   0    3136    768    768 S   0.0   0.0   0:00.00 SessionLeader
    256 root      20   0    3136   1156   1024 S   0.0   0.0   0:00.15 Relay(257)
    257 ImmunIty  20   0    8540   5248   3584 S   0.0   0.0   0:00.57 bash
    258 root      20   0    7472   3712   3456 S   0.0   0.0   0:00.01 login
    268 ImmunIty  20   0   21900  12288  10240 S   0.0   0.1   0:00.17 systemd
    270 ImmunIty  20   0   22120   3540   1920 S   0.0   0.0   0:00.00 (sd-pam)
    287 ImmunIty  20   0    5308   4480   3072 S   0.0   0.0   0:00.04 bash
   1319 ImmunIty  20   0    5976   2432   2176 T   0.0   0.0   0:00.00 less
   1349 ImmunIty  20   0    5580   1792   1792 T   0.0   0.0   0:00.00 tail
   1359 ImmunIty  20   0    5580   1792   1792 T   0.0   0.0   0:00.00 tail
   1454 ImmunIty  20   0   10300   5504   3456 R   0.0   0.0   0:00.00 top
```

29. htop (if installed)

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ htop
Could not find command-not-found database. Run 'sudo apt update' to populate it.
htop: command not found
```

30. ps aux

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  23796 13880 ?        Ss   00:25   0:01 /sbin/init
root           2  0.0  0.0   3120  1920 ?        Sl   00:25   0:00 /init
root           7  0.0  0.0   3120  1792 ?        Sl   00:25   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          44  0.0  0.1  50072 14464 ?        Ss   00:25   0:00 /usr/lib/systemd/systemd-journald
root          72  0.0  0.0  33552 10880 ?        Ss   00:25   0:02 /usr/lib/systemd/systemd-udevd
root          80  0.0  0.0   4280  2432 ?        Ss   00:25   0:00 /usr/sbin/cron -f
message+      81  0.0  0.0   7888  3840 ?        Ss   00:25   0:00 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activatio
root         114  0.0  0.0  17776  8064 ?        Ss   00:25   0:00 /usr/lib/systemd/systemd-logind
root         249  0.0  0.0   5168  2560 hvc0     Ss+  00:25   0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/u
root         250  0.0  0.0   5124  2304 tty1     Ss+  00:25   0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/u
root         255  0.0  0.0   3136   768 ?        Ss   00:25   0:00 /init
root         256  0.0  0.0   3136  1156 ?        S    00:25   0:00 /init
ImmunIty     257  0.0  0.0   8540  5248 pts/0    Ss   00:25   0:00 -bash
root         258  0.0  0.0   7472  3712 ?        Ss   00:25   0:00 login -- ImmunIty
ImmunIty     268  0.0  0.1  21900 12288 ?        Ss   00:25   0:00 /usr/lib/systemd/systemd --user
ImmunIty     270  0.0  0.0  22120  3540 ?        S    00:25   0:00 (sd-pam)
ImmunIty     287  0.0  0.0   5308  4480 pts/1    Ss+  00:25   0:00 -bash
ImmunIty    1319  0.0  0.0   5976  2432 pts/0    T    02:54   0:00 less scroll.txt
ImmunIty    1349  0.0  0.0   5580  1792 pts/0    T    02:57   0:00 tail -f scroll.txt
ImmunIty    1359  0.0  0.0   5580  1792 pts/0    T    02:57   0:00 tail -f scroll.txt
ImmunIty    1505  0.0  0.0   9280  3840 pts/0    R+   03:09   0:00 ps aux
```

Search & Filter

31. grep "text" file.txt

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ grep "ali" file.txt
I am ali
```

32. grep -i "text" file.txt

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ grep -i "ali" file.txt
I am ali
```

33. find . -name "*.txt"

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ find . -name "*.txt"
./file.txt
./scroll.txt
```

34. wc -l file.txt

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ wc -l scroll.txt
500 scroll.txt
```

35. sort file.txt

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ sort scroll.txt
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
This is a test line for scrolling
```

Networking

36. ping google.com

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ping google.com
PING google.com (142.250.187.78) 56(84) bytes of data.
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=1 ttl=111 time=87.4 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=2 ttl=111 time=65.9 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=3 ttl=111 time=60.2 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=4 ttl=111 time=68.4 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=5 ttl=111 time=62.9 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=6 ttl=111 time=62.9 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=7 ttl=111 time=55.0 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6005ms
rtt min/avg/max/mdev = 54.979/66.095/87.371/9.541 ms
```

37. curl http://example.com

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ curl http://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background
:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div>
<h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.
org/domains/example">Learn more</a></p></div></body></html>
```

38. ifconfig

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.20.121.177  netmask 255.255.240.0  broadcast 172.20.127.255
        inet6 fe80::215:5dff:fe76:88cd  prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:76:88:cd  txqueuelen 1000  (Ethernet)
        RX packets 681  bytes 67543 (65.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 69  bytes 5229 (5.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 30  bytes 3054 (2.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 30  bytes 3054 (2.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

39. netstat -tulpn

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ netstat -tulpn
(No info could be read for "-p": geteuid()=1000 but you should be root.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 10.255.255.254:53       0.0.0.0:*               LISTEN      -
udp        0      0 10.255.255.254:53       0.0.0.0:*                           -
udp        0      0 127.0.0.1:323           0.0.0.0:*                           -
udp6       0      0 ::1:323                 :::*                                -
```

40. ss -tulpn

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ss -tulpn
Netid  State      Recv-Q  Send-Q   Local Address:Port   Peer Address:Port   Process
udp    UNCONN     0       0        10.255.255.254:53    0.0.0.0:*
udp    UNCONN     0       0        127.0.0.1:323        0.0.0.0:*
udp    UNCONN     0       0        [::1]:323            [::]:*
tcp    LISTEN     0       1000     10.255.255.254:53    0.0.0.0:*
```

Package Management

41. sudo apt update

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ sudo apt update
[sudo] password for ImmunIty:
Sorry, try again.
[sudo] password for ImmunIty:
Get:1 http://kali.download/kali kali-last-snapshot InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-last-snapshot/main amd64 Packages [20.9 MB]
22% [2 Packages 7,139 kB/20.9 MB 34%]
```

42. sudo apt upgrade

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ sudo apt upgrade
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

43. sudo apt install nginx

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ sudo apt install nginx
Error: Unable to locate package nginx
```

44. sudo apt remove nginx

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ sudo apt remove nginx
Error: Unable to locate package nginx
```

Permissions

45. chmod 755 file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ chmod 755 file.txt
```

46. chown user:user file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ chown ImmunIty:ImmunIty file.txt
```

Compression

47. tar -cvf archive.tar folder/

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ tar -cvf archive.tar Test/
Test/

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ ls
archive.tar  file.txt  I  scroll.txt  Test
```

48. tar -xvf archive.tar

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ tar -xvf archive.tar
Test/
```

49. zip file.zip file

```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ zip file.zip file
Could not find command-not-found database. Run 'sudo apt update' to populate it.
zip: command not found

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ sudo apt install zip
[sudo] password for ImmunIty:
Error: Unable to locate package zip
```

50. unzip file.zip


```
┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ zip file.zip file
Could not find command-not-found database. Run 'sudo apt update' to populate it.
zip: command not found

┌──(ImmunIty㉿1mmun1ty)-[~]
└─$ sudo apt install zip
[sudo] password for ImmunIty:
Error: Unable to locate package zip
```